

Isotope Geochemistry: Epsilon Notation (ϵ_{Nd}), CHUR & Mantle-Crust Geochemical Dynamics

Comprehensive Geochemistry Guide · Authoritative References: William M. White (2005, Ch. 8) & K. C. Misra (2012, Ch. 10)

THEORETICAL FOUNDATION: THE Sm-Nd SYSTEM & EPSILON NOTATION

The Chondritic Uniform Reservoir, Derivation of the ϵ Formulation & The Terrestrial Mantle Array

1. The Sm-Nd Geochronometer & The CHUR Reference Reservoir

Samarium (Sm) and Neodymium (Nd) are adjacent Rare Earth Elements. The radioactive isotope ^{147}Sm decays to stable radiogenic ^{143}Nd by alpha (α) emission:

THE SM-ND DECAY EQUATION



- Decay constant: $\lambda_{\text{Sm}} = 6.54 \times 10^{-12} \text{ yr}^{-1}$
- Half-life: $t_{1/2} = 1.06 \times 10^{11} \text{ years}$ (106 Ga)

Normalized to the non-radiogenic reference isotope ^{144}Nd :

$$\left(\frac{^{143}\text{Nd}}{^{144}\text{Nd}} \right)_{\text{sample}}(t) = \left(\frac{^{143}\text{Nd}}{^{144}\text{Nd}} \right)_0 + \left(\frac{^{147}\text{Sm}}{^{144}\text{Nd}} \right)_{\text{sample}} \cdot (e^{\lambda t} - 1)$$

The Chondritic Uniform Reservoir (CHUR):

Because both Sm and Nd are refractory lithophile elements that condense at high temperatures in the solar nebula, their planetary ratio is not fractionated by volatilization. DePaolo & Wasserburg (1976) demonstrated that the **Bulk Silicate Earth (BSE)** possesses the exact same Sm/Nd ratio and Nd-isotopic composition as pristine chondritic meteorites. This global planetary baseline is termed **CHUR**:

- Present-day CHUR ratio: $\left(\frac{^{143}\text{Nd}}{^{144}\text{Nd}} \right)_{\text{CHUR}}(0) = 0.512638$
- Present-day CHUR parent/daughter ratio: $\left(\frac{^{147}\text{Sm}}{^{144}\text{Nd}} \right)_{\text{CHUR}} = 0.1967$

2. Epsilon (ϵ) Notation: Origin and Mathematical Definition

Because the half-life of ^{147}Sm is exceptionally long (106 Ga), radiogenic growth of ^{143}Nd is very slow. Differences in measured $^{143}\text{Nd}/^{144}\text{Nd}$ ratios between diverse terrestrial rocks occur only in the **4th and 5th decimal places** (e.g., 0.5126 vs. 0.5118).

To eliminate cumbersome decimal fractions and highlight petrogenetic deviations, Donald J. DePaolo and Gerald J. Wasserburg (1976) invented the **Epsilon (ϵ_{Nd}) parameter**, defined as the fractional deviation of a sample's Nd-isotopic ratio from that of CHUR in **parts per 10,000** (parts per myriad):

THE EPSILON NEODYMIUM (ϵ_{Nd}) FORMULATION

$$\epsilon_{\text{Nd}}(t) = \left[\frac{(^{143}\text{Nd}/^{144}\text{Nd})_{\text{sample}}(t)}{(^{143}\text{Nd}/^{144}\text{Nd})_{\text{CHUR}}(t)} - 1 \right] \times 10^4$$

For present-day rocks ($t = 0$):

$$\epsilon_{\text{Nd}}(0) = \left[\frac{(^{143}\text{Nd}/^{144}\text{Nd})_{\text{sample}}(0)}{0.512638} - 1 \right] \times 10^4$$

"Because variations in the $^{143}\text{Nd}/^{144}\text{Nd}$ ratio are small, DePaolo and Wasserburg (1976) defined the ϵ_{Nd} (epsilon) notation to express the deviation of the $^{143}\text{Nd}/^{144}\text{Nd}$ ratio of a sample from that of the Chondritic Uniform Reservoir (CHUR) in parts per 10^4 (\$10,000\$)... The beauty of this notation is that rocks derived from an undifferentiated reservoir have $\epsilon_{\text{Nd}} = 0$, while rocks from depleted mantle have positive values and ancient crustal rocks have negative values."

— William M. White, *Geochemistry* (2005), Chapter 8: Radiogenic Isotope Geochemistry, p. 325.

3. Petrogenetic Meaning: Positive vs. Negative ϵ_{Nd} Values

Because Nd has a larger ionic radius than Sm (due to the Lanthanide Contraction), Nd is *more incompatible* than Sm during mantle partial melting ($D_{\text{Nd}} < D_{\text{Sm}}$). This creates an elegant geodynamic dichotomy:

ϵ_{Nd} Value	Relative Sm/Nd Ratio	Geochemical Reservoir	Representative Geological Rock Suite
$\epsilon_{\text{Nd}} = 0$	Identical to CHUR (0.1967)	CHUR / Bulk Silicate Earth (BSE)	Primitive, undifferentiated Earth mantle; chondritic meteorites.
$\epsilon_{\text{Nd}} > 0$ (Positive) \$(+2 \text{ to } +12)\$	High Sm/Nd (\$\$ \$ \$ > 0.20\$)	Depleted Mantle (DM)	Mid-Ocean Ridge Basalts (N-MORB). Ancient melt extraction stripped more Nd than Sm into the crust, leaving a mantle residue with high Sm/Nd that evolved radiogenic, positive ϵ_{Nd} .
$\epsilon_{\text{Nd}} < 0$ (Negative) \$(-5 \text{ to } -40)\$	Low Sm/Nd (\$\$ \$ \$ < 0.18\$)	Enriched Continental Crust	Granites, Shales, Ancient Cratons. Partial melts extracted into the crust captured preferential Nd, resulting in low Sm/Nd and retarded radiogenic growth relative to CHUR.

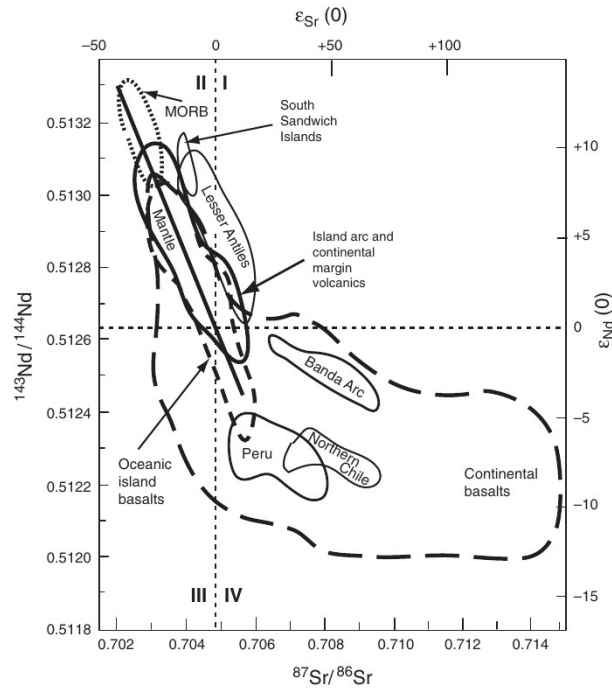


Fig. 10.15 Nd and Sr isotope compositional fields of young oceanic basalts (MORB and OIB), subduction-related basalts (island arcs formed by oceanic-oceanic and oceanic-continental subduction), and continental basalts.

Figure 1 (The Terrestrial Mantle Array): Plot of ϵ_{Nd} versus ϵ_{Sr} for oceanic and continental igneous rocks. The inverse correlation forms the renowned **Mantle Array** trending from the upper-left (Depleted Mantle: positive ϵ_{Nd} , negative ϵ_{Sr}) through the origin ($\epsilon_{\text{Nd}} = 0$, $\epsilon_{\text{Sr}} = 0$, representing BSE/CHUR) down to the lower-right (Enriched Crustal Reservoirs: negative ϵ_{Nd} , positive ϵ_{Sr}).

4. Notation Conventions Across Isotope Geochemistry

Exam candidates must never confuse epsilon notation with delta notation or other physical constants:

Symbol	Notation Name	Scale / Multiplier	Application & Domain
ϵ (The Answer)	Epsilon Notation	$\times 10^4$ (parts per 10,000)	Radiogenic Isotopes ($\epsilon_{\text{Sm-Nd}}$, $\epsilon_{\text{Lu-Hf}}$, $\epsilon_{\text{Rb-Sr}}$): Deviations from bulk planetary reservoirs (CHUR / UR).
δ	Delta Notation	$\times 10^3$ (parts per 1,000, ‰)	Stable Isotopes ($\delta^{18}\text{O}$, $\delta^{13}\text{C}$, $\delta^{34}\text{S}$, δD): Equilibrium & kinetic mass-dependent fractionation relative to SMOW, PDB, CDT.
ω	Omega	Dimensionless / Ratio	Cosmological density parameter (Ω) or specific atomic frequency. Never used for isotopic reservoir deviation.
ν	Upsilon	Dimensionless / Kinematic	Particle physics or fluid viscosity (ν). Never used in isotope geochemistry.



EXAM QUESTION & RIGOROUS OPTION-BY-OPTION DISSECTION

DGMS 2019 Examination · Official Question Statement, Verified Answer & Detailed Distracter Analysis

DGMS 2019 · QUESTION 22

Which one of the following symbols denotes the deviation of the Nd-isotopic composition of a rock from that of the chondritic uniform reservoir?

- (a) Ω (Omega)
- (b) ϵ (Epsilon)
- (c) Δ (Delta)
- (d) Υ (Upsilon)

Correct Option: (b) ϵ (Epsilon)



Verified via William M. White (Geochemistry, p. 325) & K. C. Misra (Introduction to Geochemistry, p. 248)

INCORRECT Option (a): Ω (Omega)

Detailed Refutation: The Greek letter Ω (Omega) has no role in radiogenic isotope geochemistry. In cosmology and astrophysics, Ω denotes the critical density parameter of the universe; in structural geology, it may denote angular rotation. It is never used to define deviations in radiogenic isotope ratios from CHUR.

CORRECT ANSWER Option (b): ϵ (Epsilon)

Detailed Validation: The lowercase Greek letter ϵ (Epsilon) is the universally accepted, standard symbol introduced by DePaolo & Wasserburg (1976) to denote the parts-per-ten-thousand (10^{-4}) deviation of the $^{143}\text{Nd}/^{144}\text{Nd}$ isotopic composition of a rock from that of the **Chondritic Uniform Reservoir (CHUR)**. A positive ϵ_{Nd} indicates extraction from a depleted mantle source with high Sm/Nd , while a negative ϵ_{Nd} signifies derivation from or contamination by enriched, low- Sm/Nd continental crust. This matches the exam question verbatim.

INCORRECT Option (c): δ (Delta)

Detailed Refutation: The Greek letter δ (Delta) is used exclusively for **stable isotope geochemistry** (such as $\delta^{18}\text{O}$, $\delta^{13}\text{C}$, $\delta^{34}\text{S}$, and δD), where variations are reported as parts-per-thousand (permil, $\text{‰} = 10^{-3}$) deviations from arbitrary reference standards like V-SMOW, V-PDB, or Canyon Diablo Troilite (CDT). Delta notation is never used for Nd radiogenic deviations from CHUR.

INCORRECT Option (d): ϵ (Upsilon)

Detailed Refutation: The Greek letter ϵ (Upsilon) is not used as an isotopic delta or deviation symbol anywhere in geochemistry. It serves solely as a distracter.



SYNTHESIS & HIGH-YIELD PRACTICE QUESTIONS

Curated Problems Covering Model Ages (T_{DM}), Mantle Array Trajectories & Cratonic Geochemistry

High-Yield Exam Practice Questions

Practice Q1 (Numerical / Calculation): Evaluating ϵ_{Nd}

A modern basalt from the East Pacific Rise has a measured $^{143}\text{Nd}/^{144}\text{Nd} = 0.513150$. Given that present-day CHUR has $^{143}\text{Nd}/^{144}\text{Nd} = 0.512638$, calculate $\epsilon_{\text{Nd}}(0)$ and interpret its mantle provenance.

Solution & Rationale:

$$\epsilon_{\text{Nd}}(0) = \left[\frac{0.513150}{0.512638} - 1 \right] \times 10^4 = [1.00099875 - 1] \times 10^4 = \mathbf{+9.99} \approx \mathbf{+10.0}$$

A value of $\epsilon_{\text{Nd}} = +10.0$ indicates derivation from the **Depleted Mantle (DM)** reservoir, which underwent ancient melt depletion that elevated its Sm/Nd ratio relative to CHUR.

Practice Q2 (Geochronology): Nd Model Age (T_{DM}) Definition

What is a Neodymium Depleted Mantle Model Age (T_{DM}), and why is it referred to as a "crustal residence age"?

Answer: The T_{DM} age is the calculated time in the past when the Nd-isotopic growth curve of a crustal rock intersected the Depleted Mantle evolution curve. It records the time when the rock's chemical precursors were **originally extracted from the mantle** to form continental crust, irrespective of later metamorphic or magmatic remelting events.

Practice Q3 (Scale Multiplier): Parts Per Ten Thousand vs. Permil

Why did DePaolo and Wasserburg select a multiplier of 10^4 (parts per ten thousand) for epsilon notation instead of 10^3 (permil, used in delta notation)?

Answer: Because the fractional differences in $^{143}\text{Nd}/^{144}\text{Nd}$ caused by Sm-Nd decay are approximately an order of magnitude smaller than mass-dependent stable isotope fractionation. A multiplier of 10^4 scales the subtle differences in the 4th and 5th decimal places into convenient, whole-integer numbers (e.g., $+10$ or -15) easily analyzed by geochemists.

Practice Q4 (Complementary Systems): Lu-Hf and Epsilon Hafnium (ϵ_{Hf})

Which other radiogenic isotope system is defined using an identical epsilon notation relative to a chondritic reservoir, and how does its behavior correlate with ϵ_{Nd} ?

Answer: The **Lutetium-Hafnium (Lu-Hf) system** (^{176}Lu to ^{176}Hf), expressed as ϵ_{Hf} relative to CHUR. Because Lu-Hf behaves identically to Sm-Nd during mantle melting ($D_{\text{Hf}} < D_{\text{Lu}}$), ϵ_{Hf} correlates positively with ϵ_{Nd} along the "terrestrial mantle-crust array" ($\epsilon_{\text{Hf}} \approx 1.4 \epsilon_{\text{Nd}} + 3$).

Practice Q5 (Crustal Provenance): Archean Craton Signatures

A granitic pluton intruded during the Proterozoic (1.8 Ga) displays an initial $\epsilon_{\text{Nd}}(1.8 \text{ Ga}) = -12.5$. What does this imply about the magma's origin?

Answer: It proves that the granite was not derived from juvenile Proterozoic mantle; rather, it formed by the **remelting or heavy assimilation of ancient Archean continental crust** that had existed for at least 1 billion years prior with low Sm/Nd , allowing its ϵ_{Nd} to decay to highly negative values.

Practice Q6 (Sediment Resistance): Sm-Nd vs. Weathering

Why do Nd isotopes in clastic sedimentary rocks (e.g., shales and turbidites) accurately record the average crustal age of an entire eroded continent, whereas Sr isotopes do not?

Answer: Rare Earth Elements like Sm and Nd are insoluble and immobile during surface weathering and diagenesis, preserving their pristine parent-daughter ratio. Strontium (Sr), being a fluid-mobile alkaline earth element, is easily leached and exchanged with seawater and groundwaters, corrupting its original isotopic signature.

Practice Q7 (Mantle End-Members): Zindler and Hart (1986) Taxonomy

In the isotope taxonomy of oceanic basalts, what mantle component is characterized by extremely high $^{206}\text{Pb}/^{204}\text{Pb}$ but moderately positive ϵ_{Nd} ?

Answer: **HIMU** (High- μ , where $\mu = ^{238}\text{U}/^{204}\text{Pb}$). HIMU represents ancient subducted oceanic crust that was dehydrated (stripping lead relative to uranium) and recycled through the deep mantle for $1.5\text{--}2.0 \text{ Ga}$.

Practice Q8 (Initial Ratio Evolution): Slopes of ϵ_{Nd} vs. Time

On a plot of ϵ_{Nd} versus geologic time (4.55 Ga to Present), what is the slope of the CHUR reference reservoir line?

Answer: **Zero (horizontal line at $\epsilon_{\text{Nd}} = 0$)**. Because ϵ_{Nd} is defined by normalize-comparing a sample to CHUR at that exact same time t , CHUR by mathematical definition always has $\epsilon_{\text{Nd}}(t) \equiv 0$ at all times in Earth's history.

Comprehensive Isotope Ratio Notation Matrix

Notation Symbol	Isotope System	Reference Baseline	Multiplier	Geological Usage
ϵ_{Nd} \$ (The Answer)	^{147}Sm to ^{143}Nd	CHUR (Chondritic Uniform Reservoir)	$\times 10^4$ (parts per \$10,000)	Mantle depletion, continental crust extraction, crustal recycling.
ϵ_{Hf} \$	^{176}Lu to ^{176}Hf	CHUR	$\times 10^4$ (parts per \$10,000)	Zircon provenance, crustal growth through geologic time.
ϵ_{Sr} \$	^{87}Rb to ^{87}Sr	UR (Uniform Reservoir)	$\times 10^4$ (parts per \$10,000)	Mantle array correlation with ϵ_{Nd} .
$\delta^{18}\text{O}$, $\delta^{13}\text{C}$	Stable Isotopes	V-SMOW, V- PDB	$\times 10^3$ (parts per \$1,000, ‰)	Paleothermometry, paleoclimate, fluid- rock interaction.